Project-X: Forcing Visual Grounding

Causal-Aware Few-Shot Learning in Vision-Language Models
via Counterfactual Retrieval and Attention Modulation

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Outline

Workflow

Causal retrieval $\rightarrow$ k-shot OpenFlamingo $\rightarrow$ LoRA grounding $\rightarrow$ measure accuracy $+$ causal image use

Problem Statement

VLM In-Context Learning is fragile in three ways:

1. Modality imbalance

Text-driven ICL; images largely ignored (weak cross-attention).

2. Spurious retrieval

RICES $=$ passive similarity $\Rightarrow$ *non-causal* demos.

3. Many-shot paradox

More shots $\Rightarrow$ worse; optimum $=$ narrow few-shot window.

Goal: force *causal visual grounding*.

State of the Art

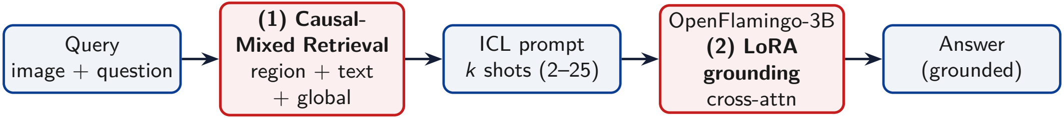
Prior work

- Flamingo / OpenFlamingo — gated cross-attn ICL
- RICES — CLIP-similarity retrieval
- ICL analyses — visual modality under-used
- Counterfactual CIR — counterfactuals help
- Many-shot studies — more can hurt

Gap addressed

- Prior: retrieval *or* architecture
- Ours: causal retrieval + grounding loss
- Validated by *causal reliance*, not only accuracy

Proposed Method — Dual Novelty



(1) Causal-Mixed Retrieval

- Tri-weighted: region+text+global
- Class-diversity filter
- Bounded shot window

(2) Grounding Penalty

- Counterfactual grounding loss
- LoRA on gated cross-attn
- Uses the *correct* image

Method (1) — Causal-Mixed Retrieval

Score each demo vs. query — region + text + global:

$$\mathrm{score}(c) = 0.4\ \cos\left(f_q^{\mathrm{loc}}, f_c^{\mathrm{loc}}\right) + 0.3\ \cos\left(f_q^{\mathrm{txt}}, f_c^{\mathrm{txt}}\right) + 0.3\ \cos\left(f_q^{\mathrm{glob}}, f_c^{\mathrm{glob}}\right)$$

- CLIP ViT-B/32 features; region = bbox crop (else global)
- Class-diversity filter, then back-fill
- Shots clamped $[2, 32]$; tested $k \in \{2, 5, 15, 25\}$

Why it helps

Region $\Rightarrow$ local causal match; diversity $\Rightarrow$ no near-duplicates.

Method (2) — Cross-Modal Attention-Forcing Penalty

LoRA on gated cross-attn; $\ell_b = \mathrm{NLL}(\text{real})$, $\ell_c = \mathrm{NLL}(\text{wrong})$:

$$\mathcal{L} = \underbrace{\ell_b}_{\text{task}} + \lambda_g \underbrace{\mathrm{ReLU}(m - (\ell_c - \ell_b))}_{\text{wrong img worse by } m} + \lambda_e \underbrace{H(\text{attn}_{\text{answer}})}_{\text{sharpen attention}}$$

- Grounding: wrong image must be worse by margin m
- Entropy: sharpen attention on answer tokens
- PEFT only: **3.24M / 2.69B** $\approx$ **0.12%** params

Dataset

Train — grounding adapter

VQAv2 (val stream, $N = 2000$) — general VQA, no benchmark leakage

Zero-/few-shot evaluation (held-out):

GQA — testdev balanced

- Compositional, low language bias
- Tests grounding (spatial)
- `lmms-lab/GQA`

AwA2 — Animals w/ Attributes

- 50 classes, 85 attributes
- Tests causal retrieval
- `KRadim/AwA-Pose-Lite`

$n = 150$ queries / ablation / shot level

Experimental Setup

Model & training

- OpenFlamingo-3B: CLIP ViT-L/14 + MPT-1B, cross-attn every layer.
- LoRA: r=16, α=32, dropout 0.05 on `to_q/to_kv/to_out`.
- AdamW, lr 2×10^{-5}, wd 0.01, 1 epoch, 150 steps.
- λ_g=0.5, margin m=0.5, λ_e=0.05.
- Seed 42; eval in fp16; CLIP ViT-B/32 retriever.

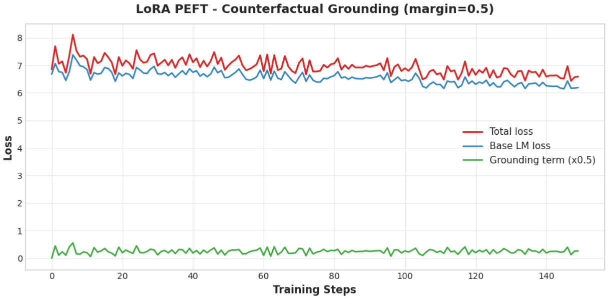
2×2 factorial ablation

	adapter **OFF**	adapter **ON**
random retr.	Baseline	Ablation 3
causal retr.	Ablation 1	Ablation 2

Ab1 = retrieval; **Ab3** = grounding; **Ab2** = full.

Metrics: EM (= GQA accuracy), F1, confidence (geo-mean answer-token prob.); image reliance: ΔNLL = NLL(blind)−NLL(real), span-JSD, blindfold gap, flip rate; GQA answer-distribution χ^2.

Training — Counterfactual Grounding Curve



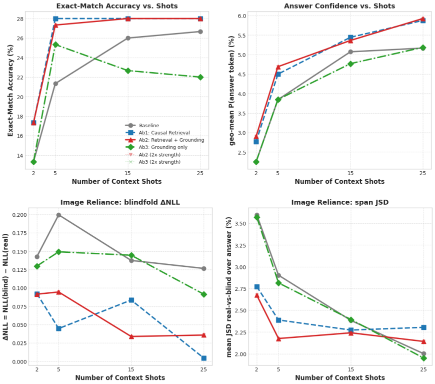
- Base LM loss **6.68** $\rightarrow$ **6.19**: adapter does *not* damage the language model
- Grounding term stays active (mean ≈ 0.45): the margin "wrong image must be worse" keeps being enforced
- Attention entropy **3.48** $\rightarrow$ **2.68** (-23%): cross-attention concentrates on fewer visual tokens

$\Rightarrow$ all three signals move as designed: the model is pushed to use the *correct* image at no LM cost

Model Evaluation — GQA (Exact-Match %)

Config	2	5	15	25
Baseline	13.3	21.3	26.0	26.7
Ab1 Causal retr.	**17.3**	**28.0**	**28.0**	**28.0**
Ab2 Retr.+Ground	**17.3**	27.3	**28.0**	**28.0**
Ab3 Ground only	13.3	25.3	22.7	22.0

- EM **+4.0 to +6.7** pts at *every k*: consistent, not one lucky setting
- F1 21.3 → **32.2**, confidence 3.8 → **4.7** % (5-shot): more certain, not an exact-match artefact
- ΔNLL *drops* (+0.20 → +0.04): gains come from better demos, **not** more image use ⇒ see diagnostics

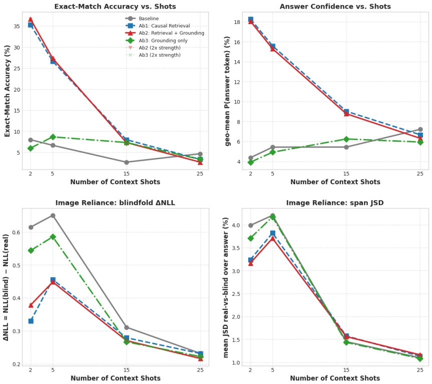


EM · confidence · ΔNLL · span-JSD vs. shots (GQA, n=150)

Model Evaluation — AwA2 (Exact-Match %)

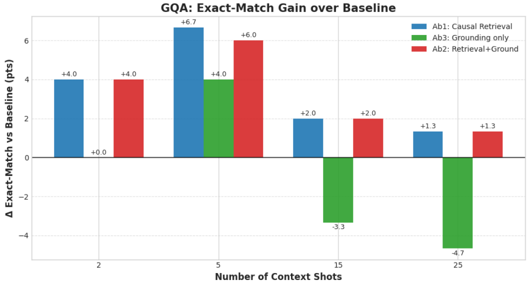
Config	2	5	15	25
Baseline	8.0	6.7	2.7	4.7
Ab1 Causal retr.	**35.3**	**26.7**	**8.0**	3.3
Ab2 Retr.+Ground	**36.7**	**27.3**	7.3	2.7
Ab3 Ground only	6.0	8.7	7.3	3.3

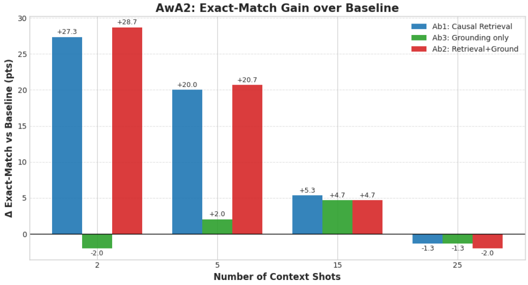
- 2-shot: 8.0 → **36.7** EM (4.6×); confidence 4.4 → **18.3** % (4×): confident answers, not guesses
- ΔNLL stays high (+0.33 to +0.61): here the image *is* genuinely used (unlike GQA)
- **Many-shot collapse**: all configs ≤ 4.7 at 25 shots; χ^2 0.9 → **8.8**: degenerate answer distribution (context copying)



EM · confidence · ΔNLL · span-JSD vs. shots (AwA2, n=150)

Gain over Baseline — which component drives accuracy?





Reading the evidence

- Ab1 $\approx$ Ab2 $\gg$ Ab3 on both benchmarks $\Rightarrow$ **causal retrieval is the accuracy driver**
- Grounding *without* causal demos turns **negative** on GQA at 15–25 shots (-3.3, -4.7): forced image attention hurts when demos are uninformative
- Gains shrink with k: bounded few-shot window confirmed

Grounding Diagnostics — Does the model use the image?

Blindfold (5-shot): swap the query image. Gap = EM(real)−EM(mismatch); flip = % of answers that change. Accuracy alone cannot prove grounding — these two metrics can.

AwA2

Config	real	mism.	**gap**	flip %
Baseline	8.3	0.0	+8.3	53.3
Ab1	26.7	8.3	+18.3	55.0
Ab2	30.0	5.0	**+25.0**	55.0
Ab3	6.7	0.0	+6.7	53.3

GQA

Config	real	mism.	**gap**	flip %
Baseline	18.3	13.3	+5.0	36.7
Ab1	28.3	26.7	+1.7	38.3
Ab2	25.0	23.3	+1.7	**41.7**
Ab3	18.3	13.3	+5.0	33.3

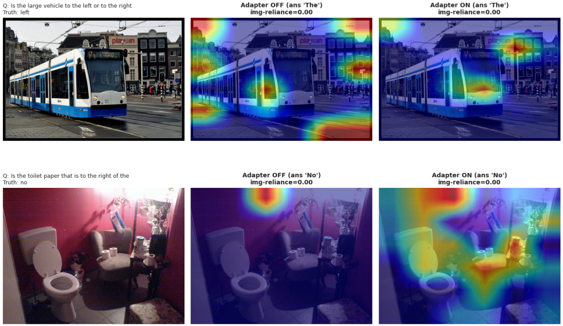
- AwA2: full method **triples** the gap (+8.3 → +25.0) *and* cuts wrong-image EM 8.3 → 5.0: correct answers now *require* the right image — causal grounding, not luck
- GQA: gap shrinks but flips *rise* (36.7 → 41.7 %): answers do move with the image; language priors often rescue EM anyway

Grounding Diagnostics — Classification & Saliency

AwA2 likelihood classification
(2-shot, $n = 20$, rank by log-lik.)

Config	top-1	top-5
Baseline	25.0	45.0
Ab1	**45.0**	**65.0**
Ab2	**45.0**	**65.0**
Ab3	10.0	50.0

- Top-1 nearly **doubles** ($25 \to 45\,\%$): gain holds under likelihood ranking too
- Ab3 top-1 **10 %**: grounding needs causal demos $\Rightarrow$ the components are complementary



Occlusion saliency (GQA): adapter ON redistributes saliency, but reliance stays ≈ 0 — matches the small GQA blindfold gap (language priors).

Conclusions & Future Work

What worked — and the evidence

- **Causal retrieval drives accuracy**: **+28.7** EM AwA2 (2-shot, 4.6×), **+6.7** GQA; confidence up to 4×; confirmed under likelihood ranking (top-1 25 → 45 %)
- **Grounding adapter drives causality**: blindfold gap **+8.3** → **+25.0**, wrong-image wins cut 8.3 → 5.0 — at zero accuracy cost and 0.12 % trained params
- **Many-shot paradox reproduced and diagnosed**: collapse beyond 5 shots with χ^2 0.9 → 8.8 (context copying)

Limitations & future work

- Grounding adds reliance, not accuracy (Ab2 ≈ Ab1); occlusion reliance on GQA stays ≈ 0
- $n = 150$/config; add GQA Validity / Plausibility (official choices file) and significance tests
- Scale n and backbone; learn the retrieval weights end-to-end

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